IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Applied Data Science Capstone | Falcon 9 first-stage landing prediction



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Code, queries and repository

GitHub repository: github.com/<your-username>/<your-repo>

Executive Summary

Methodology

- **Collect:** SpaceX REST API + Wikipedia web scraping
- **Wrangle:** clean, impute, define a binary landing Class
- **Explore:** visual EDA and SQL on 101 Falcon launches
- **Visualise:** Folium site maps, Plotly Dash dashboard
- **Predict:** Logistic Regression, SVM, Decision Tree, KNN tuned with 10-fold GridSearchCV

Key results

61 of 101

launches landed the first stage (2010-2020)

0% to 88%

yearly landing success, 2014 to 2020

77%

best site success rate: KSC LC-39A

71%

best model test accuracy (Decision Tree)

Booster generation and payload explain most of the variation.

Introduction

Background

SpaceX reuses the Falcon 9 first stage, which is why it can advertise launches at a fraction of the price of competitors.

$62M
SpaceX Falcon 9 price

$165M+
typical competitor price

If we can predict whether the first stage lands, we can estimate the real cost of a launch, and a new entrant (Space Y) can bid more intelligently.

Questions to answer

1. Which factors drive landing success: launch site, orbit, payload, booster version, experience?
2. How has landing success evolved over time?
3. Can a classifier predict the landing outcome well enough to inform pricing?

Section 1

Methodology



Methodology

Collect	Wrangle	Explore	Visualise	Predict
SpaceX API Web scraping	Missing values Binary Class	Matplotlib / Seaborn SQL	Folium maps Plotly Dash	LR, SVM, Tree, KNN GridSearchCV

Data science methodology: from raw JSON and HTML to a deployable landing predictor

Reproducible

Every step lives in a notebook in the GitHub repository

Traceable

Each result slide links back to the notebook that produced it

Comparable

Four classifiers tuned with identical splits and folds

Data Collection

SpaceX REST API

- Launch, rocket, launchpad, payload and core endpoints
- JSON to pandas DataFrame

Wikipedia scraping

- Falcon 9 launch records page
- BeautifulSoup parses the HTML tables

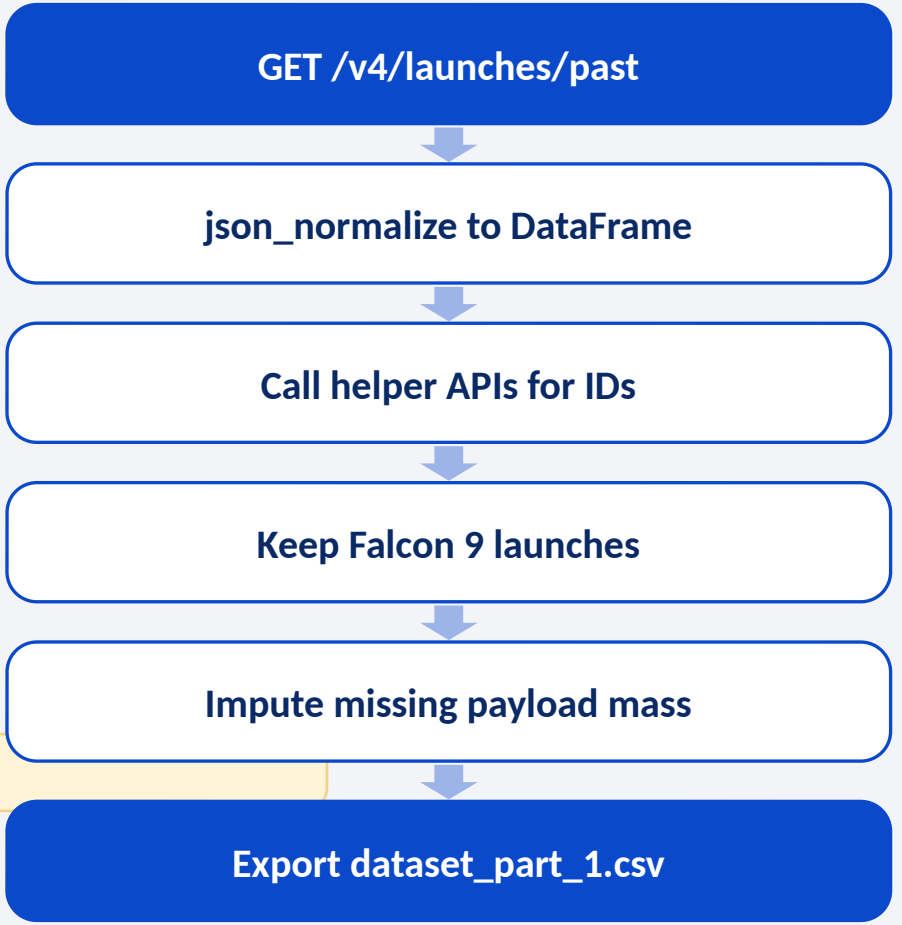
Combined launch dataset

FlightNumber	Date	BoosterVersion
PayloadMass	Orbit	LaunchSite
Outcome	Flights	GridFins
Reused	Legs	LandingPad
Block	ReusedCount	Serial
Longitude	Latitude	

Falcon 9 launches only, one row per launch, 17 descriptive fields.

Data Collection: SpaceX API

- **Request** launch history from the SpaceX v4 REST API
- **Resolve** rocket, launchpad, payload and core IDs with helper functions
- **Filter** Falcon 9 only, single-core, single-payload launches
- **Clean** replace missing PayloadMass with the mean

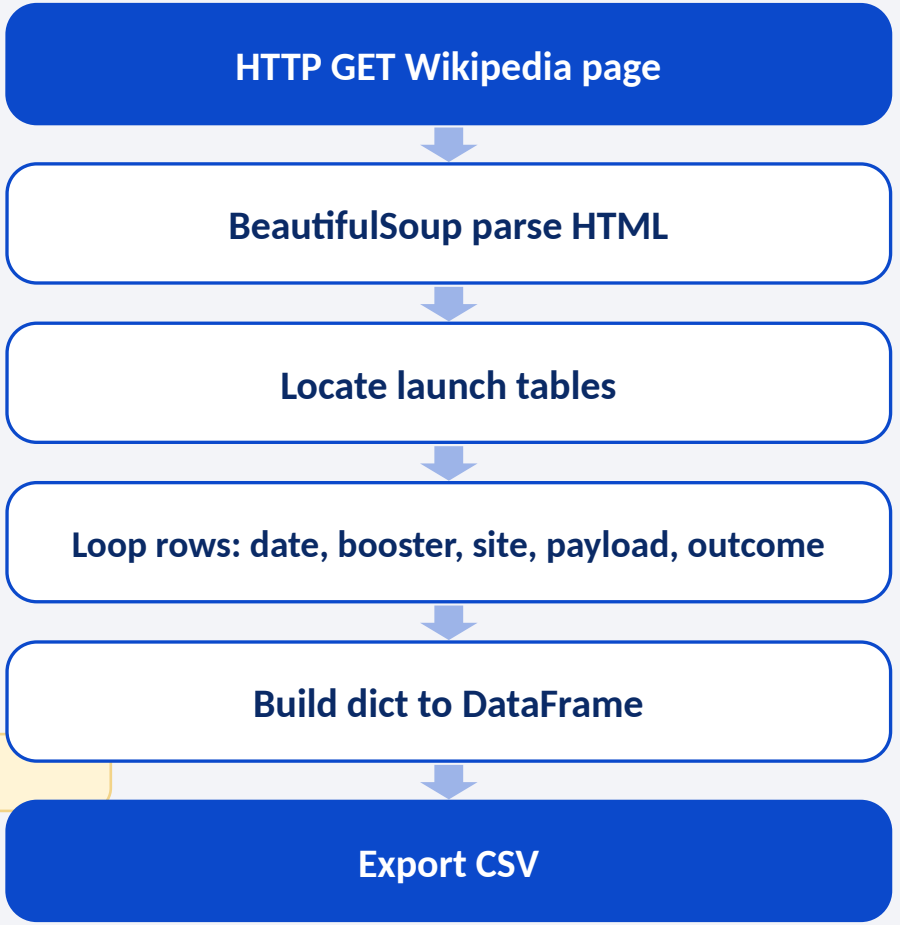


GitHub: github.com/<your-username>/<your-repo>/jupyter-labs-spacex-data-collection-api.ipynb

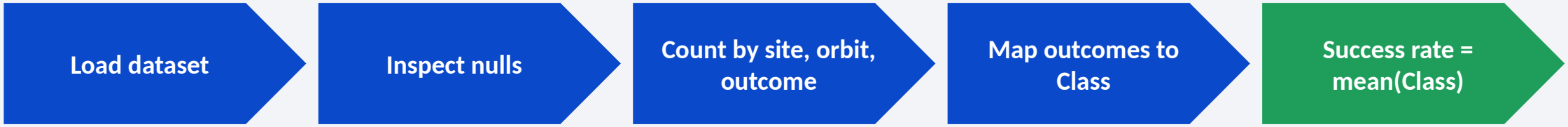
Data Collection: Web Scraping

- **Request** the Falcon 9 launch records page with an HTTP GET
- **Parse** HTML with BeautifulSoup
- **Extract** column names and every launch row from the launch tables
- **Assemble** a dictionary into a pandas DataFrame

GitHub: github.com/<your-username>/<your-repo>/jupyter-labs-webscraping.ipynb



Data Wrangling



Landing outcome to training label

Outcome	Meaning	Class
True ASDS / RTLS / Ocean	Landed (drone ship, ground pad, ocean)	1
False ASDS / RTLS / Ocean	Landing attempted but failed	0
None ASDS / None None	No landing attempted	0

The Class column becomes the target variable for every model.

Checks performed

- Missing PayloadMass imputed with the mean
- LandingPad nulls kept: no pad means no landing attempt
- Launch counts per site, orbit and mission outcome

GitHub: github.com/<your-username>/<your-repo>/labs-jupyter-spacex-Data wrangling.ipynb

EDA with Data Visualization

Chart	Variables	Why this chart
Scatter	Flight number vs launch site	Does experience at a pad improve outcomes?
Scatter	Payload vs launch site	Do pads handle different payload ranges?
Bar	Success rate by orbit	Which orbits are easiest to land from?
Scatter	Flight number vs orbit	Which orbits were flown early or late?
Scatter	Payload vs orbit	How does mass relate to orbit and outcome?
Line	Yearly success rate	Is there a clear learning curve over time?

GitHub: github.com/<your-username>/<your-repo>/edadataviz.ipynb

Green = landed, red = failed in every chart

EDA with SQL

1. Unique launch sites
2. Five launches from sites starting with CCA
3. Total payload carried for NASA (CRS)
4. Average payload of booster F9 v1.1
5. First successful ground-pad landing date
6. Drone-ship successes with payload 4,000-6,000 kg
7. Count of success and failure mission outcomes
8. Boosters that carried the maximum payload
9. 2015 drone-ship landing failures
10. Landing outcomes ranked, 2010-06-04 to 2017-03-20

GitHub: github.com/<your-username>/<your-repo>/jupyter-labs-eda-sql-coursera_sqllite.ipynb

SQLite database queried from Jupyter

Build an Interactive Map with Folium

Map object	What it shows	Why it was added
Circle + Marker	Each launch site, with the NASA JSC reference point	Locate sites and compare their surroundings
MarkerCluster	Every launch, green = success and red = failure	Spot which sites win or lose most often
MousePosition	Live latitude and longitude on hover	Read coordinates to measure distances
PolyLine + distance label	Site to coastline, railway, highway and city	Quantify how close each site is to infrastructure

GitHub: github.com/<your-username>/<your-repo>/lab_jupyter_launch_site_location.ipynb

Sites cluster near coastlines (safe over-water trajectories) and far from cities.

Build a Dashboard with Plotly Dash

Inputs	Callbacks	Outputs
• Launch-site dropdown • Payload range slider (0-10,000 kg)	• Filter the data frame • Return updated figures	• Success pie chart • Payload vs outcome scatter, coloured by booster

Why these interactions

- Dropdown: compare a single site with all sites without writing new code
- Slider: test whether payload mass changes success, booster by booster

GitHub: github.com/<your-username>/<your-repo>/spacex_dash_app.py

Predictive Analysis (Classification)

One-hot encode → Standardise → 80/20 split → GridSearchCV, 10 folds → Test accuracy and confusion matrix

Features (15)

- Flight number, payload mass, booster block
- Orbit (one-hot, 8 categories)
- Launch site (one-hot, 4 sites)
- Target: Class (1 = first stage landed)

Models and search spaces

- **Logistic Regression:** C
- **SVM:** kernel, C, gamma
- **Decision Tree:** criterion, depth, leaf and split sizes
- **KNN:** k = 1-10, Manhattan or Euclidean

GitHub: github.com/<your-username>/<your-repo>/SpaceX_Machine_Learning_Prediction_Part_5.ipynb

Results

1

Exploratory data analysis

Learning curve: success rose from 0% (2014) to 88% (2020). Newer boosters land far more often.

2

Interactive analytics

Folium maps and a Plotly Dash dashboard show KSC LC-39A as the strongest site.

3

Predictive analysis

A tuned Decision Tree is best (test accuracy 71%), though the test set is small.

Plus two new insights on booster era and payload headroom, then a pricing playbook for Space Y.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

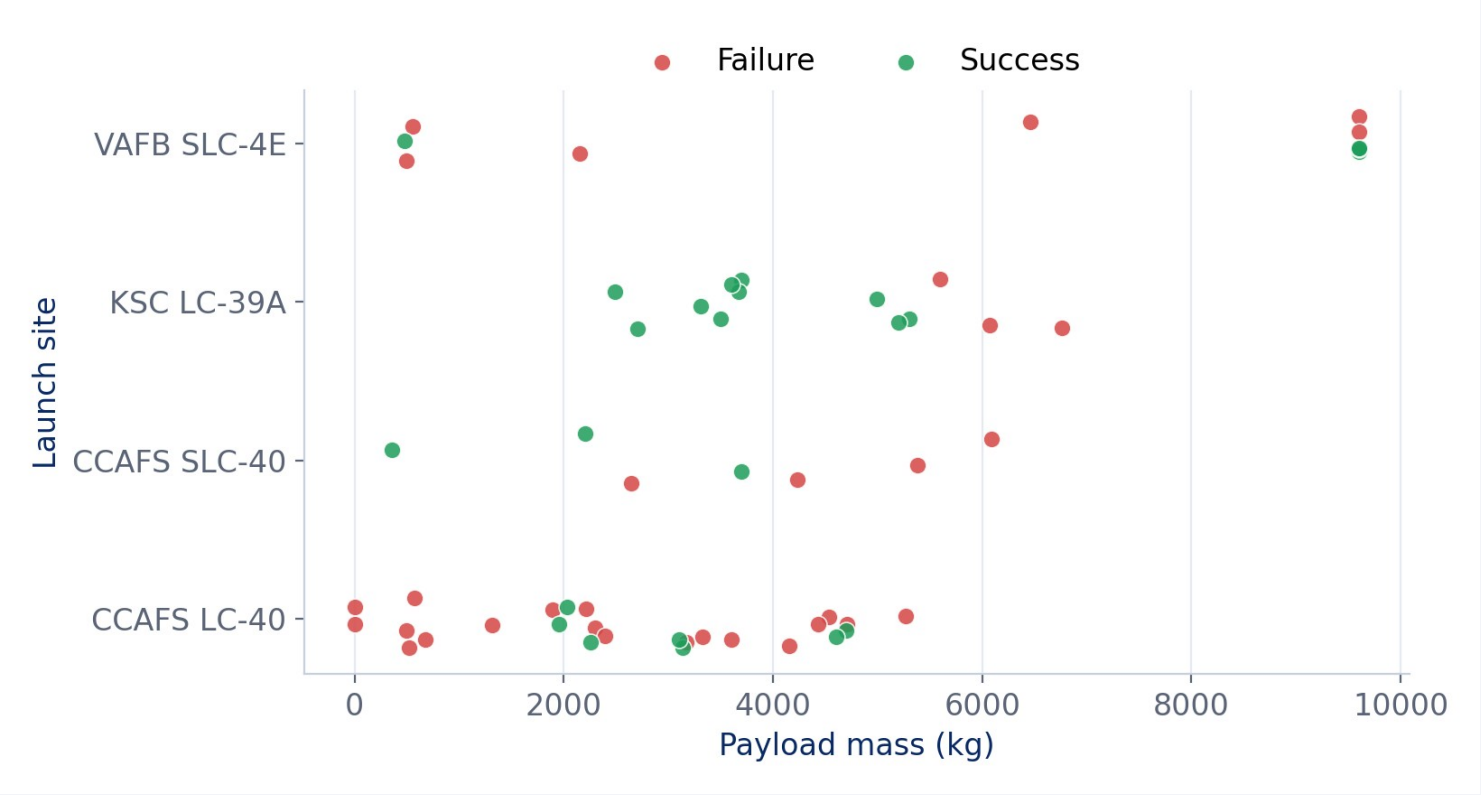


What it shows

- CCAFS LC-40 hosted flights 1-28, mostly failed early on
- KSC LC-39A opens at flight 30 and mostly lands
- Later flights land more often overall

Source: spacex_launch_dash.csv (56 landing-attempt launches)

Payload vs. Launch Site

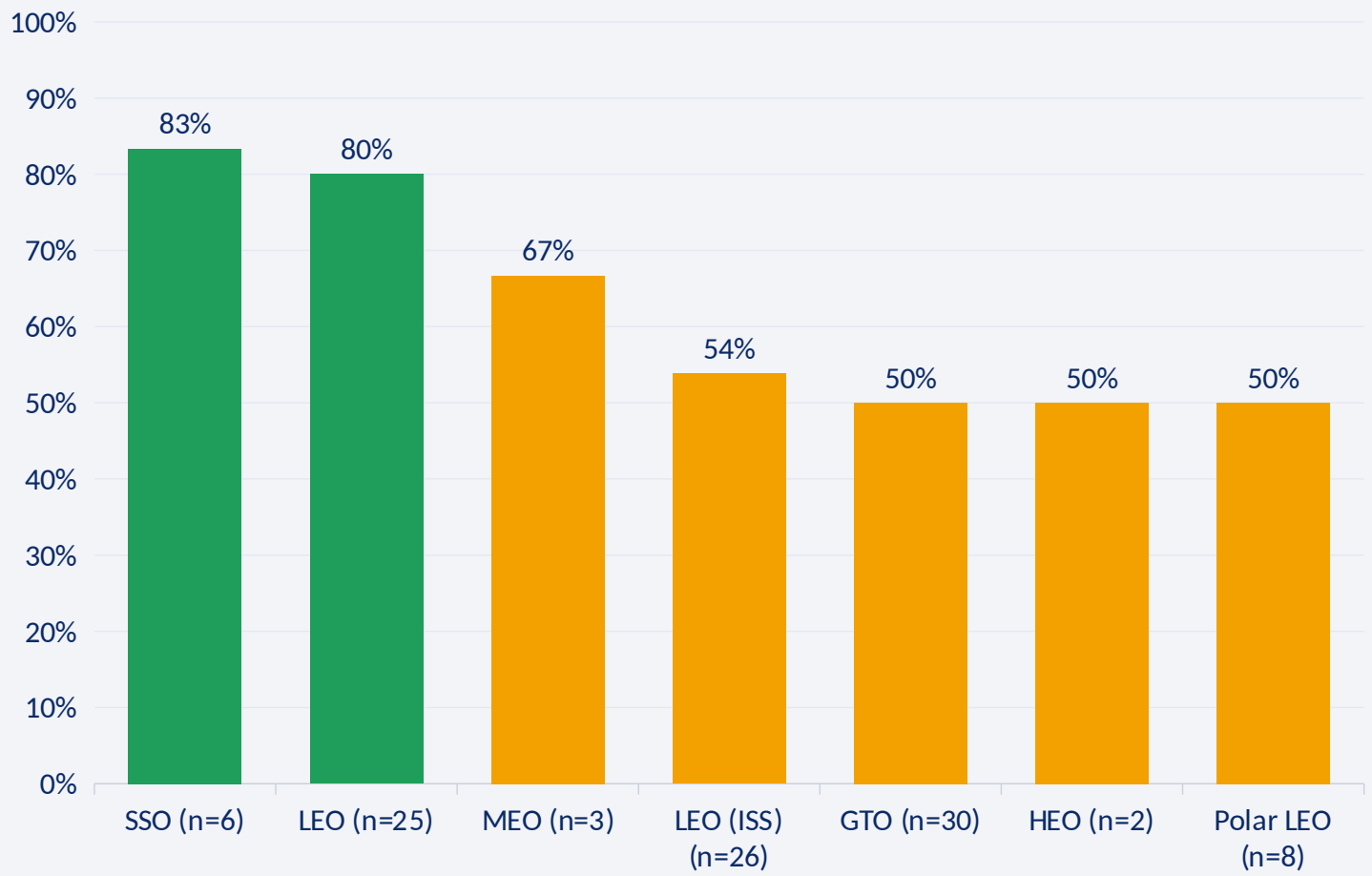


What it shows

- VAFB SLC-4E flies payloads up to 9,600 kg
- KSC LC-39A median payload: 3,697 kg
- CCAFS LC-40 spans 0-5,271 kg
- Heavier loads mostly fly from KSC and VAFB

Source: spacex_launch_dash.csv

Success Rate vs. Orbit Type

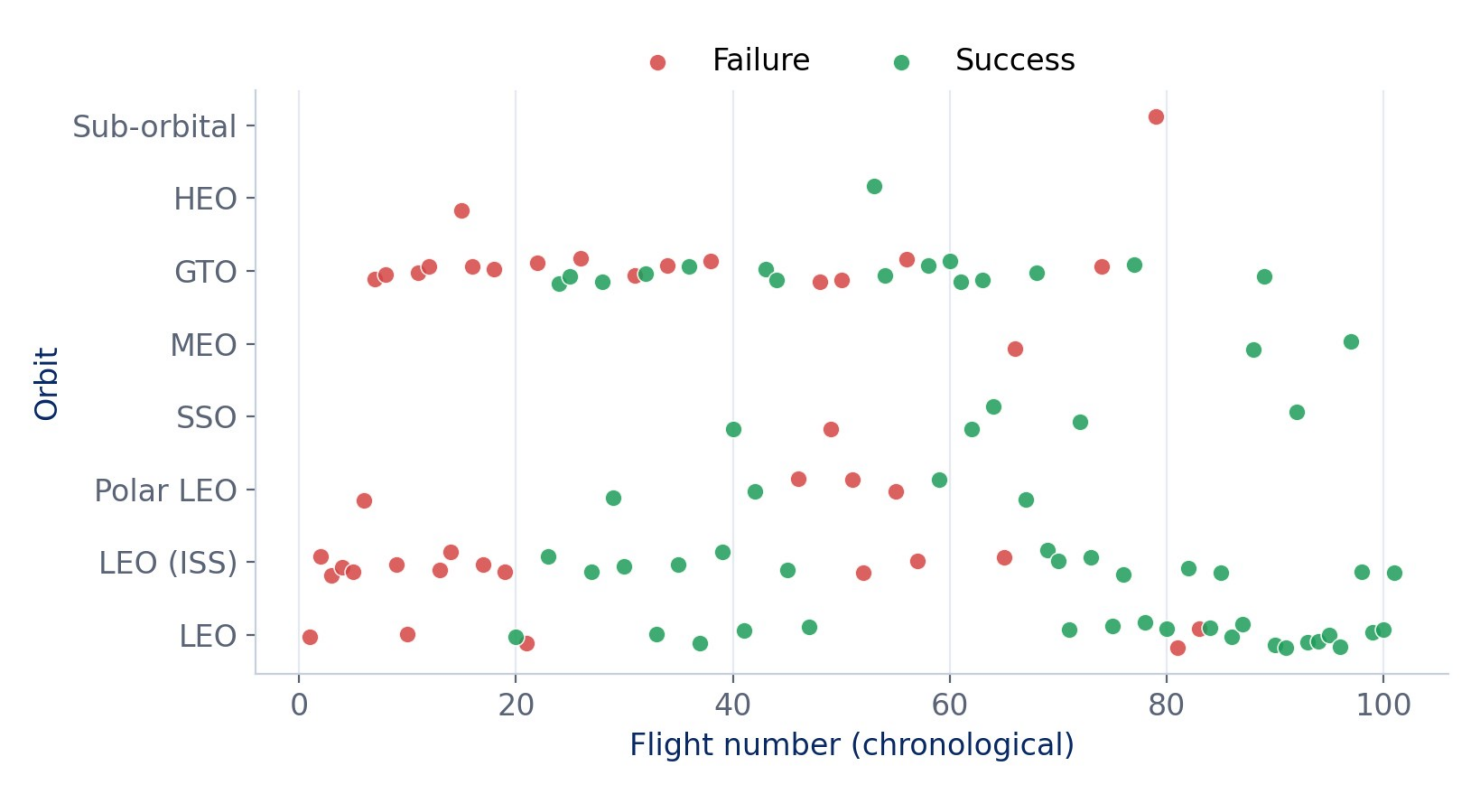


What it shows

- SSO (83%) and LEO (80%) land most often
- GTO and ISS missions land about half the time
- Orbits with n below 3 are excluded: too few launches to trust

Source: SpaceX launch table, landing success = Landing_Outcome starts with Success

Flight Number vs. Orbit Type

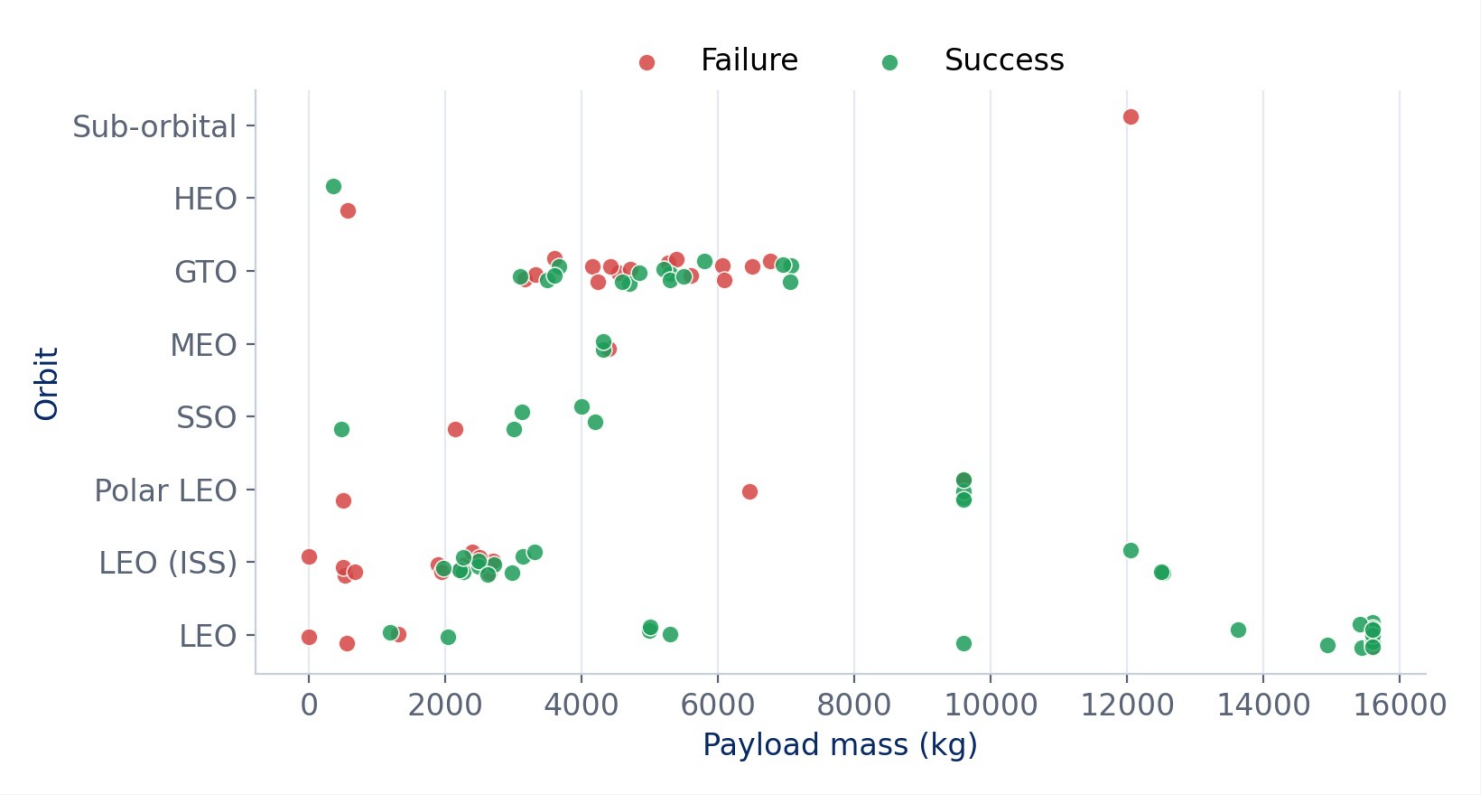


What it shows

- LEO and ISS missions run through the whole timeline
- GTO first appears at flight 7 and recurs throughout
- Newer flights succeed across orbits

Source: SpaceX launch table, flight number = chronological order

Payload vs. Orbit Type

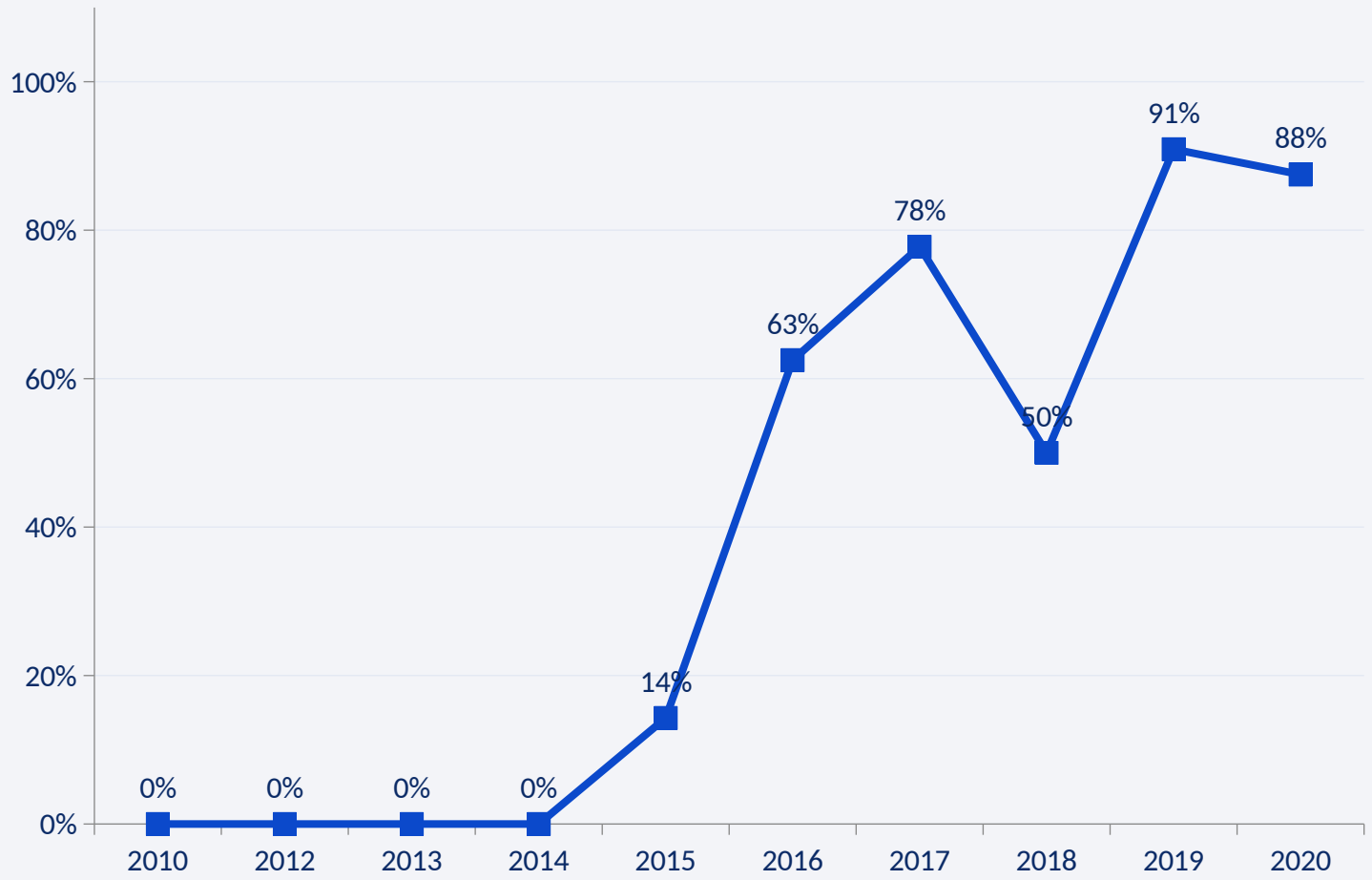


What it shows

- GTO payloads cluster at 3,100-7,075 kg
- LEO carries the heaviest loads, up to 15,600 kg
- Heavy LEO launches mostly land

Source: SpaceX launch table

Launch Success Yearly Trend



What it shows

- No successful landing before 2015
- Big jump in 2016 as drone-ship landings arrived
- 88% success in 2020
- 2018 dip: heavy missions flown without recovery

Source: SpaceX launch table (101 launches, 2010-2020)

All Launch Site Names

SQL

```
SELECT DISTINCT Launch_Site
FROM SPACEXTBL;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Explanation

DISTINCT collapses the table to the four launch sites used: three on the Florida coast (CCAFS LC-40, CCAFS SLC-40, KSC LC-39A) and one in California (VAFB SLC-4E).

Launch Site Names Begin with 'CCA'

SQL

```
SELECT *
FROM SPACEXTBL
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

Date	Booster	Site	kg	Landing
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	0	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	0	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	525	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	500	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	677	No attempt

Explanation

LIKE with a wildcard finds both Cape Canaveral pads. The first five launches (2010-2013) were early F9 v1.0 flights, none of which landed.

Total Payload Mass

SQL

```
SELECT SUM(PAYLOAD_MASS__KG_)
FROM SPACEXTBL
WHERE Customer = 'NASA (CRS)';
```

45,596 kg

total payload for NASA (CRS)

Explanation

Boosters have carried 45,596 kg of cargo for NASA Commercial Resupply Services, the supply missions to the International Space Station.

Average Payload Mass by F9 v1.1

SQL

```
SELECT AVG(PAYLOAD_MASS__KG_)
FROM SPACEXTBL
WHERE Booster_Version = 'F9 v1.1';
```

2,928.4 kg

average payload, booster F9 v1.1

Explanation

The F9 v1.1 booster averaged about 2,928 kg per launch, well below later boosters that carried up to 15,600 kg.

First Successful Ground Landing Date

SQL

```
SELECT MIN(Date)
FROM SPACEXTBL
WHERE Landing_Outcome =
  'Success (ground pad)';
```

2015-12-22

first ground-pad landing

Explanation

The first ground-pad landing took place on 2015-12-22, on the OG2 Mission 2 flight carrying 11 Orbcomm satellites. It marked the start of routine booster recovery.

Drone Ship Successes, Payload 4,000-6,000 kg

SQL

```
SELECT Booster_Version
FROM SPACEXTBL
WHERE Landing_Outcome =
  'Success (drone ship)'
AND PAYLOAD_MASS__KG_ > 4000
AND PAYLOAD_MASS__KG_ < 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Explanation

Four Falcon 9 Full Thrust boosters landed on a drone ship while carrying mid-weight payloads, evidence that heavier launches can still be recovered.

Successful and Failed Mission Outcomes

SQL

```
SELECT Mission_Outcome,
  COUNT(*) AS total
FROM SPACEXTBL
GROUP BY Mission_Outcome;
```

Outcome	Count
Success	100
Failure (in flight)	1

Explanation

100 of 101 missions succeeded: more than 99%. Mission success is separate from booster landing: a mission can succeed even when the booster is not recovered.

Boosters Carried Maximum Payload

SQL

```
SELECT DISTINCT Booster_Version
FROM SPACEXTBL
WHERE PAYLOAD_MASS__KG_ = (
  SELECT MAX(PAYLOAD_MASS__KG_)
  FROM SPACEXTBL);
```

Booster_Version (15,600 kg)
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4

Explanation

A subquery finds the maximum payload, 15,600 kg. 12 different Block 5 boosters carried it, so the heaviest loads are routine for the newest hardware. Six shown.

2015 Launch Records

SQL

```
SELECT Date, Booster_Version,
  Launch_Site
FROM SPACEXTBL
WHERE Landing_Outcome =
  'Failure (drone ship)'
AND substr(Date,1,4) = '2015';
```

Date	Booster	Site
2015-01-10	F9 v1.1 B1012	CCAFS LC-40
2015-04-14	F9 v1.1 B1015	CCAFS LC-40

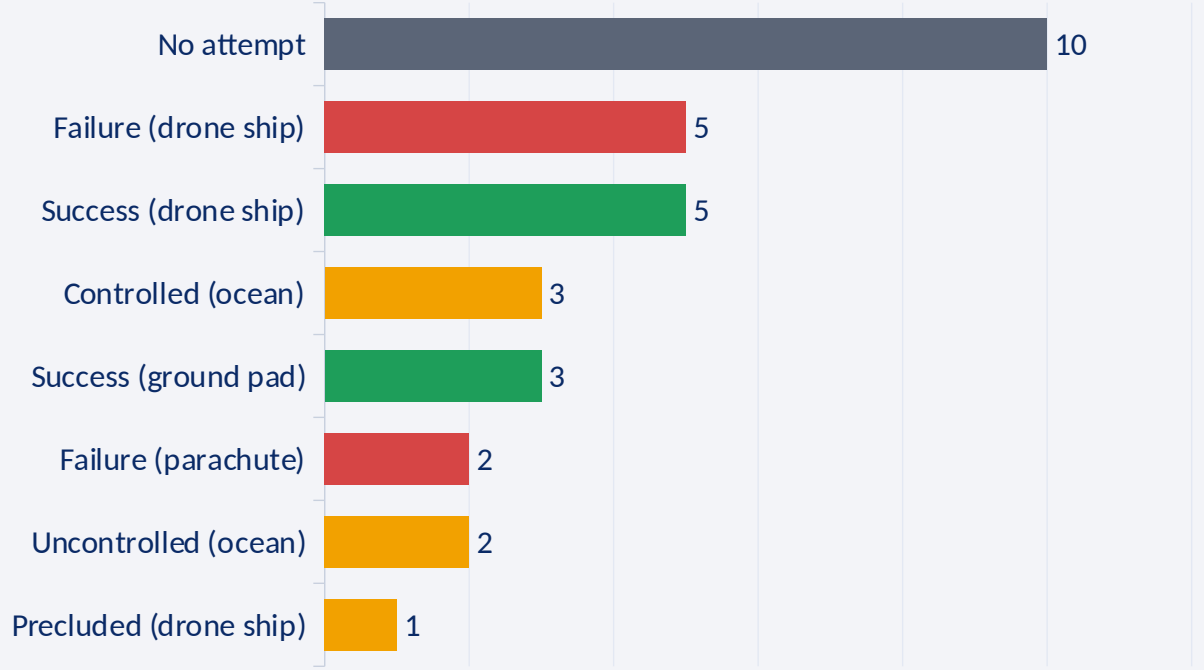
Explanation

Both 2015 drone-ship failures were F9 v1.1 boosters launched from CCAFS LC-40, early attempts at a technique that later became routine.

Rank Landing Outcomes, 2010-06-04 to 2017-03-20

SQL

```
SELECT Landing_Outcome,
  COUNT(*) AS n
FROM SPACEXTBL
WHERE Date BETWEEN
  "2010-06-04" AND "2017-03-20"
GROUP BY Landing_Outcome
ORDER BY n DESC;
```



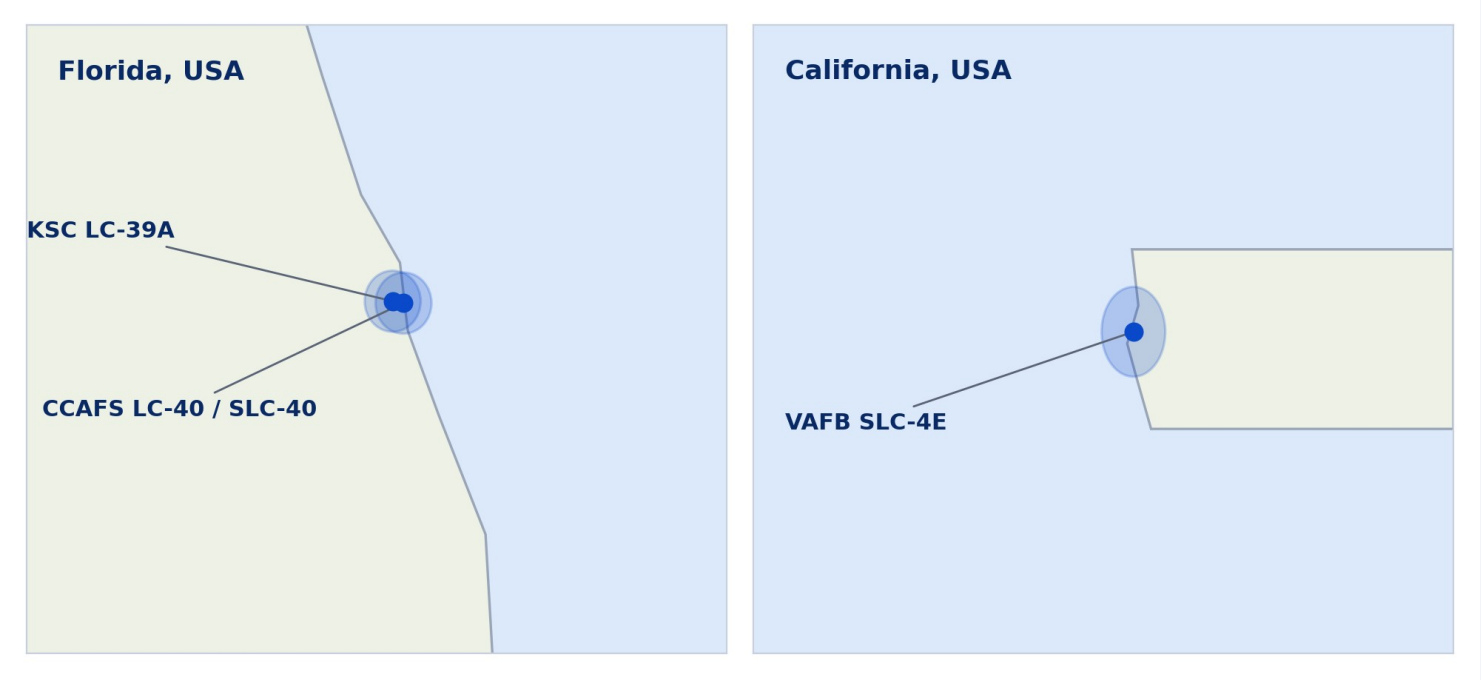
"No attempt" leads (10), then drone-ship successes and failures tie at 5. Landing was still experimental in this window.



Section 3

Launch Sites Proximities Analysis

All Launch Sites Sit Near the Coast

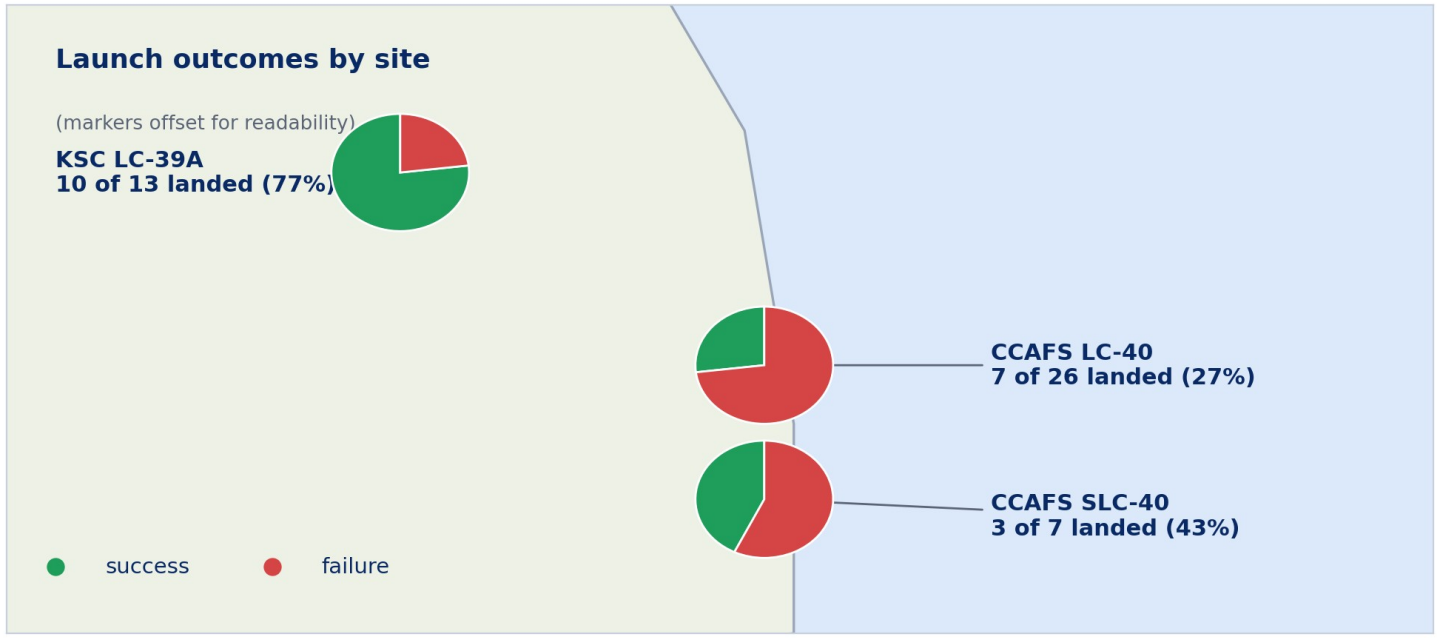


Findings

- Three sites on Florida's Atlantic coast, one in California
- Low latitudes give rockets extra speed from Earth's spin
- Coastal sites keep trajectories over water

Schematic reproduction of the Folium output, built from launch-site coordinates

Launch Outcomes by Site

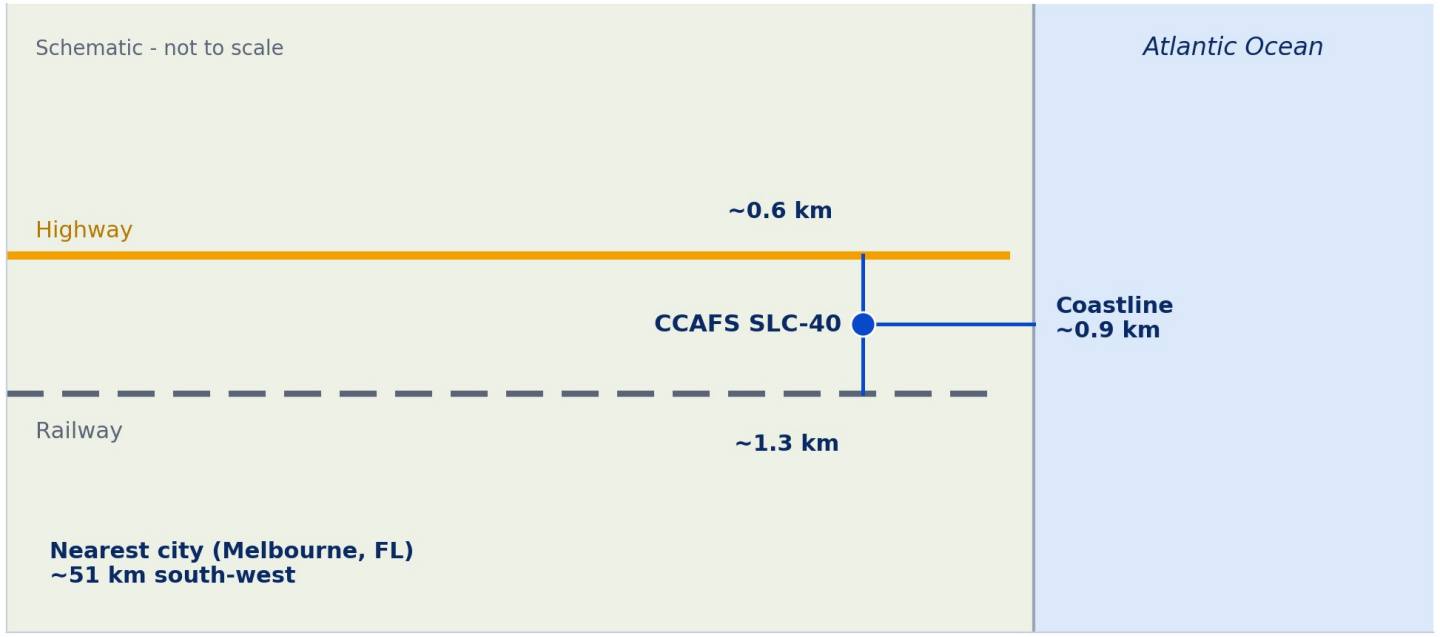


Findings

- Green = landed, red = failed
- KSC LC-39A: 10 of 13 landed (77%)
- CCAFS LC-40: 7 of 26 (27%), the oldest and busiest pad
- Newer pads show the best records

Schematic reproduction of the Folium output, built from launch-site coordinates

Launch Site Proximities: CCAFS SLC-40



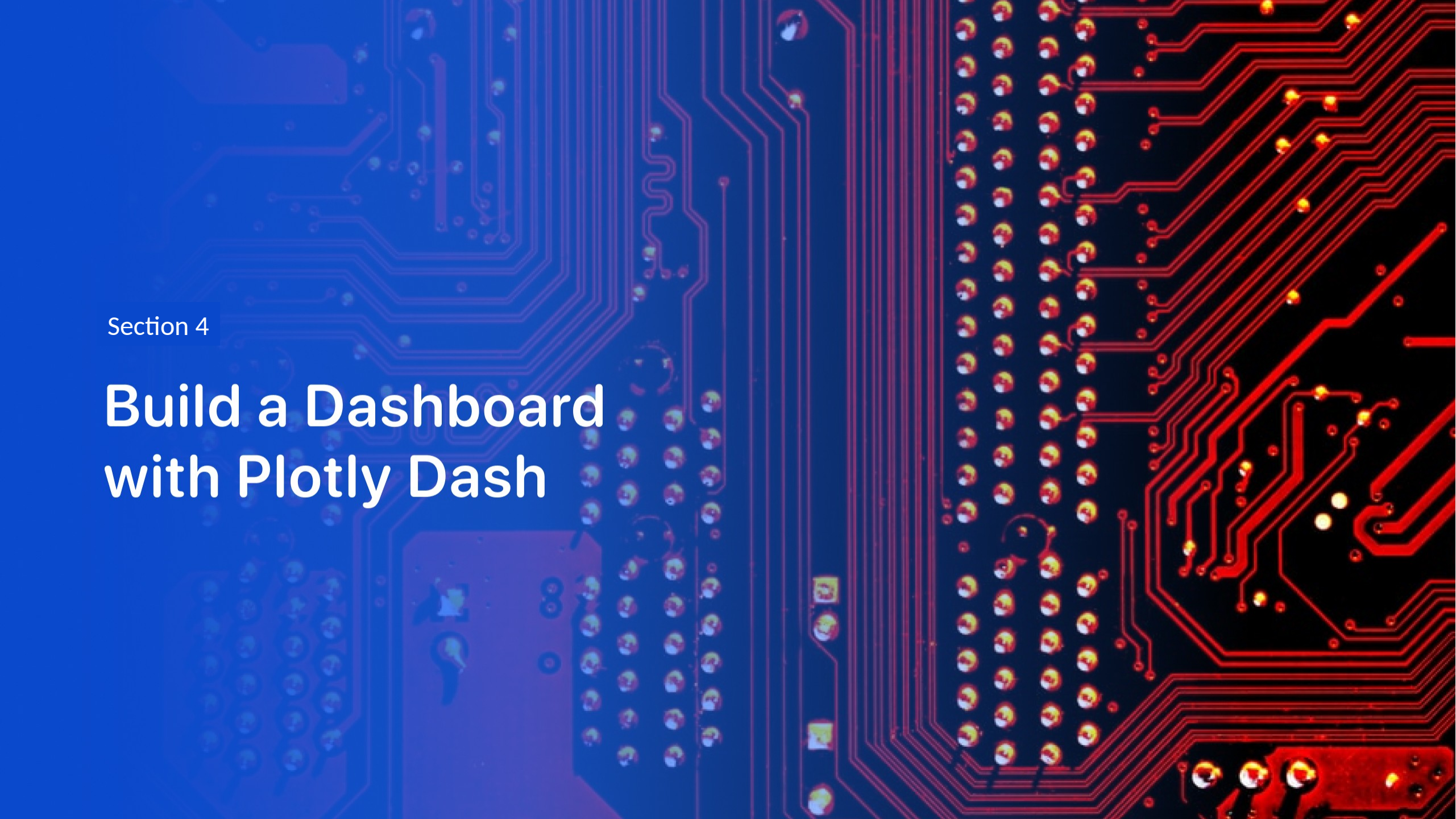
Findings

- Coastline ~0.9 km, highway ~0.6 km, railway ~1.3 km
- Nearest city ~51 km away
- Close to transport links for logistics, far from people for safety

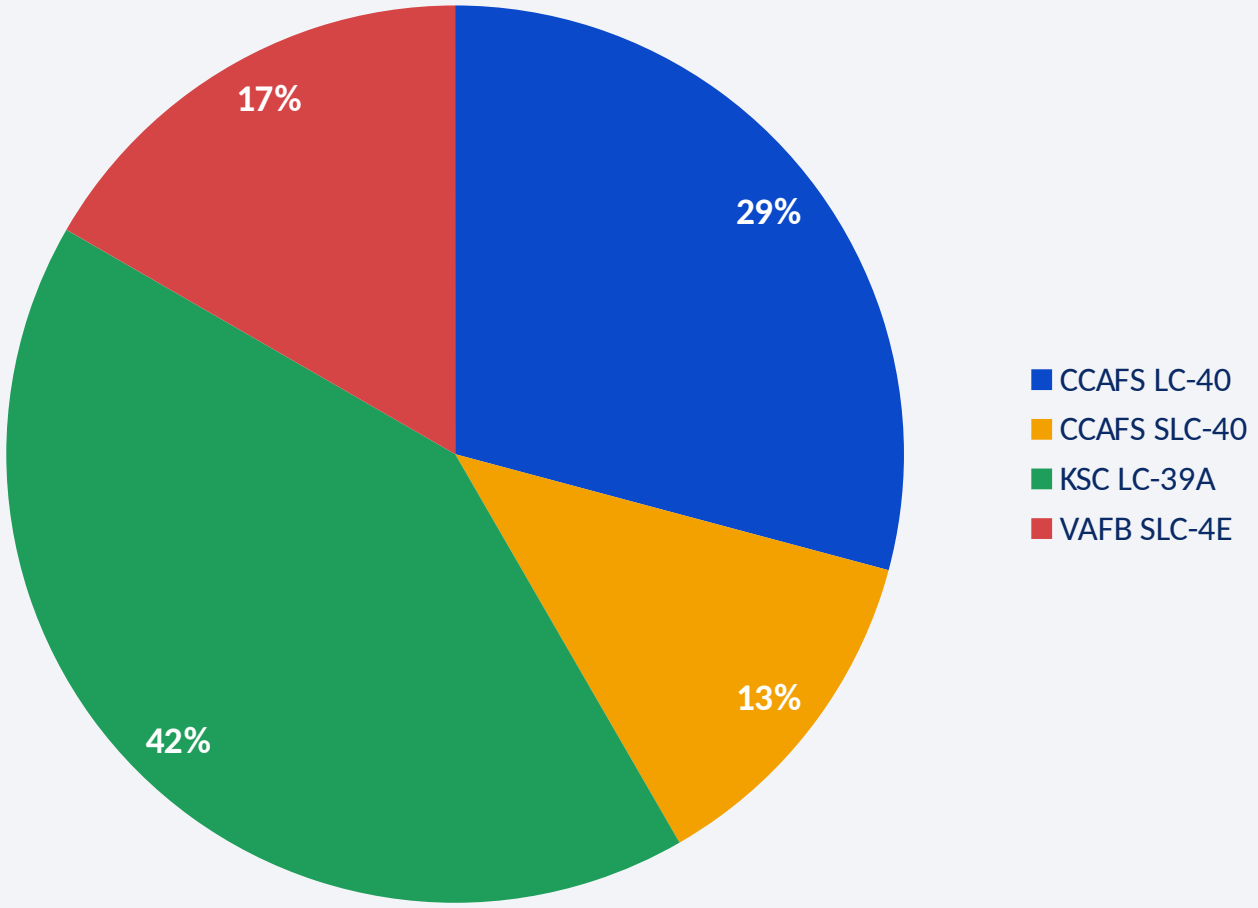
Schematic, distances approximate (Folium distance lab); not to scale

Section 4

Build a Dashboard with Plotly Dash



Launch Success Count by Site

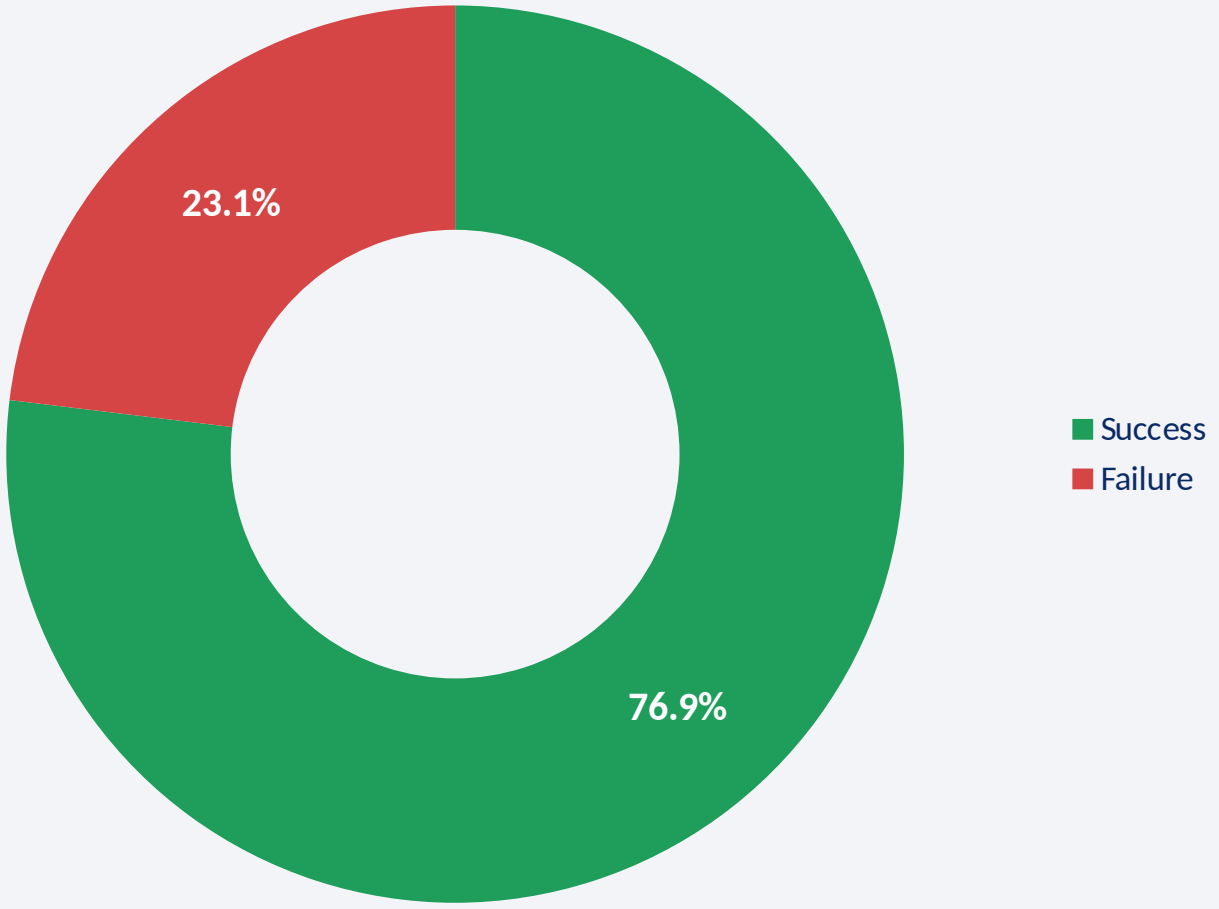


Findings

- KSC LC-39A holds 41.7% of all successful landings
- 24 landings across 56 attempts
- Dropdown = All Sites

Dashboard view reproduced from spacex_launch_dash.csv

Highest Success Ratio: KSC LC-39A

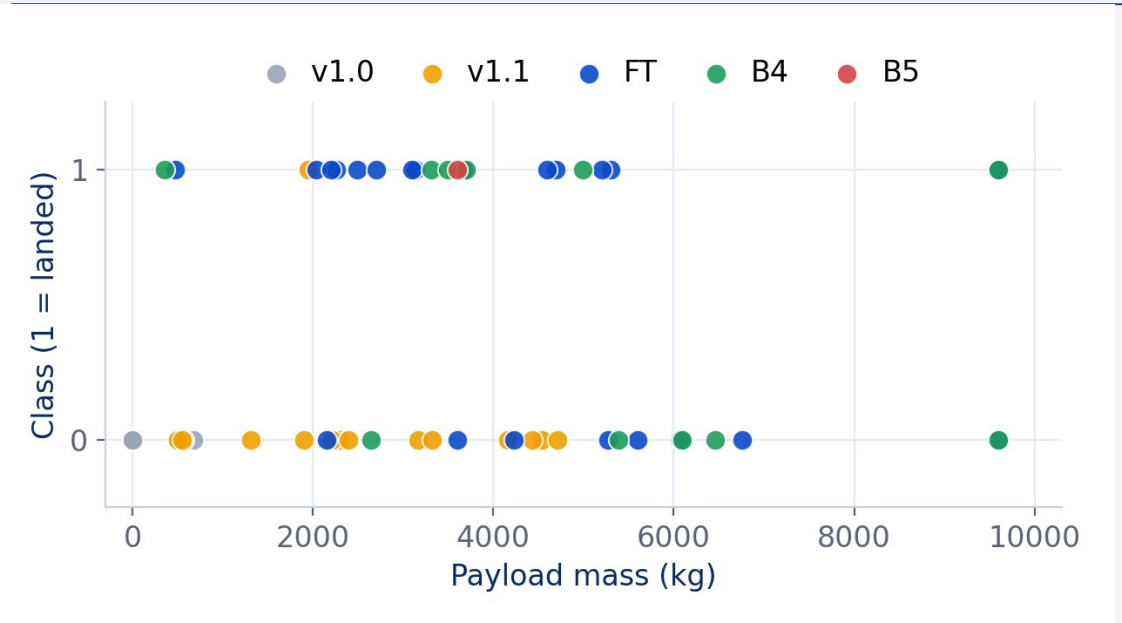


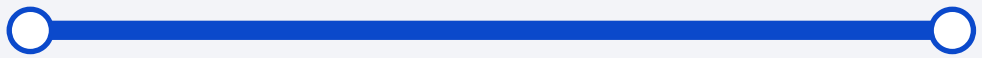
Findings

- 10 successes, 3 failures: a 76.9% ratio
- Next best: CCAFS SLC-40 at 42.9%
- All KSC LC-39A launches used modern boosters

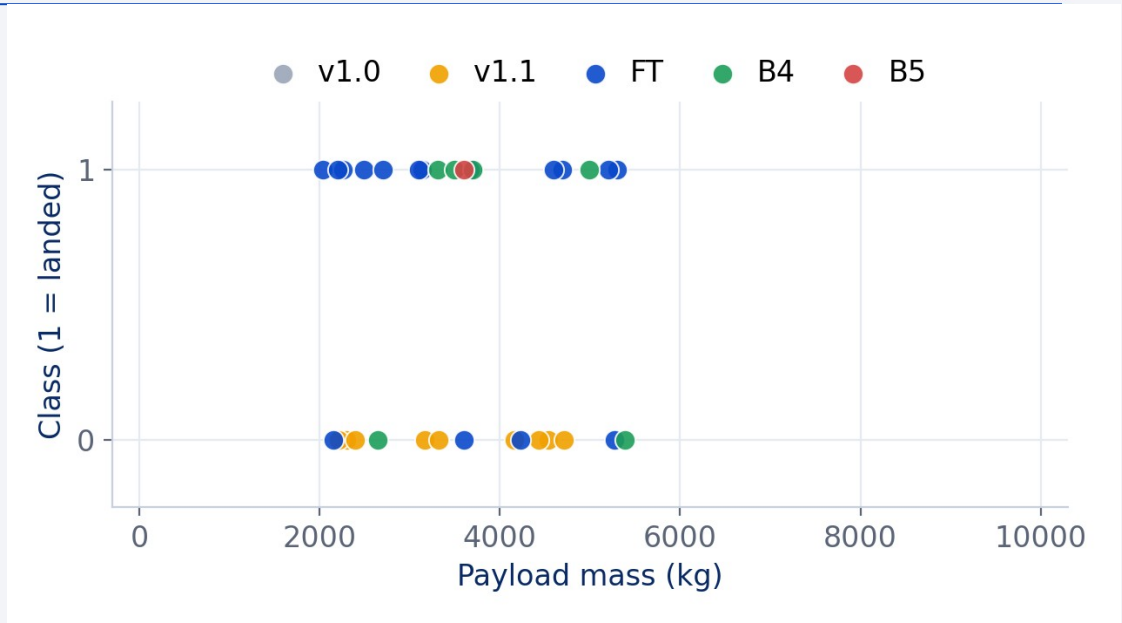
Dashboard view reproduced from spacex_launch_dash.csv

Payload vs. Launch Outcome





Slider: 0 to 10,000 kg





Slider: 2,000 to 5,500 kg

- 2,000-4,000 kg has the best success rate: 62%
- Below 2,000 kg: 23%, mostly old v1.0/v1.1 boosters
- FT boosters (blue) dominate the successes
- B4 landed 6 of 11; the single B5 landed

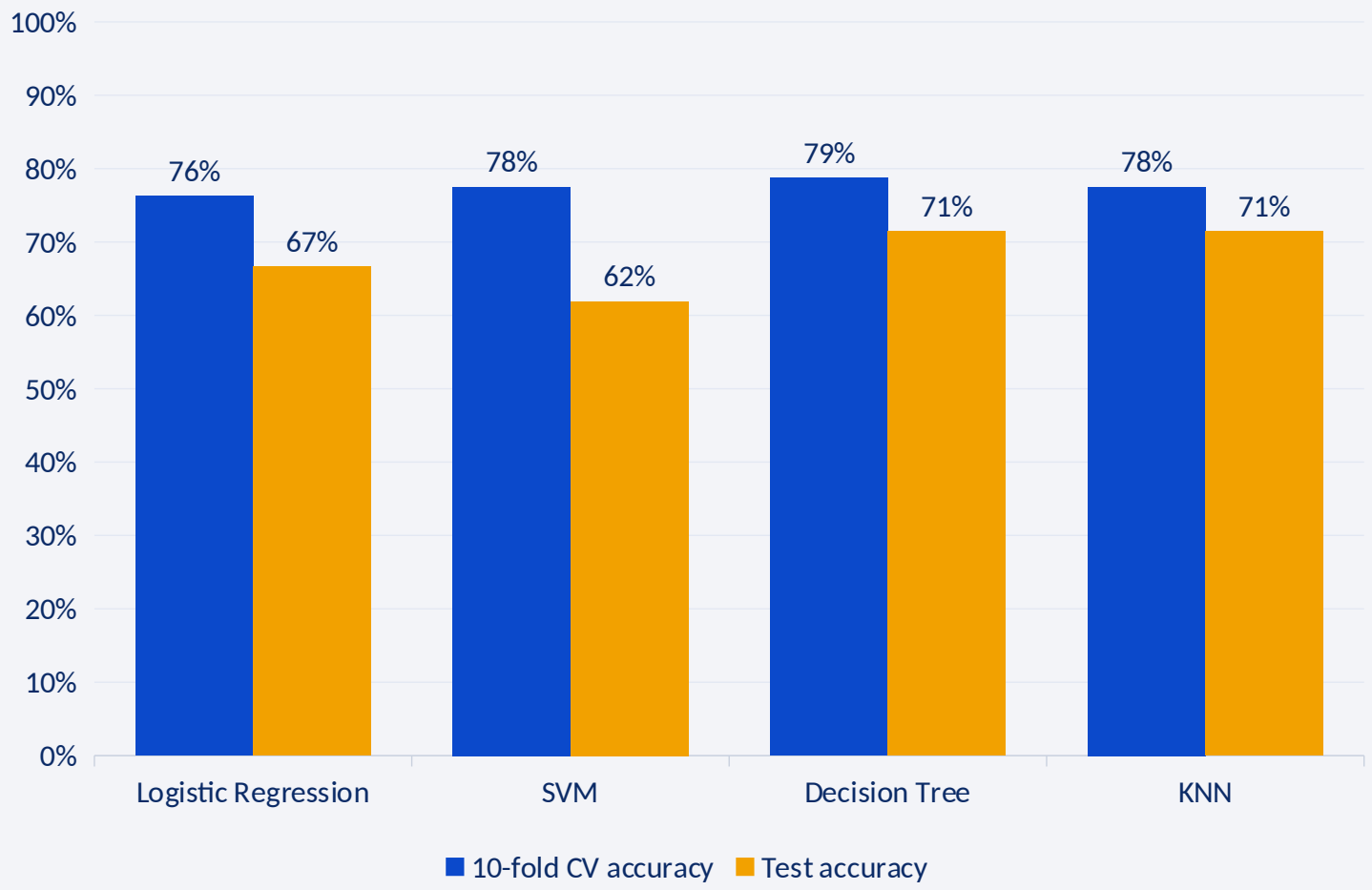
Dashboard view reproduced from spacex_launch_dash.csv; slider shown at two settings

Section 5

Predictive Analysis (Classification)



Classification Accuracy

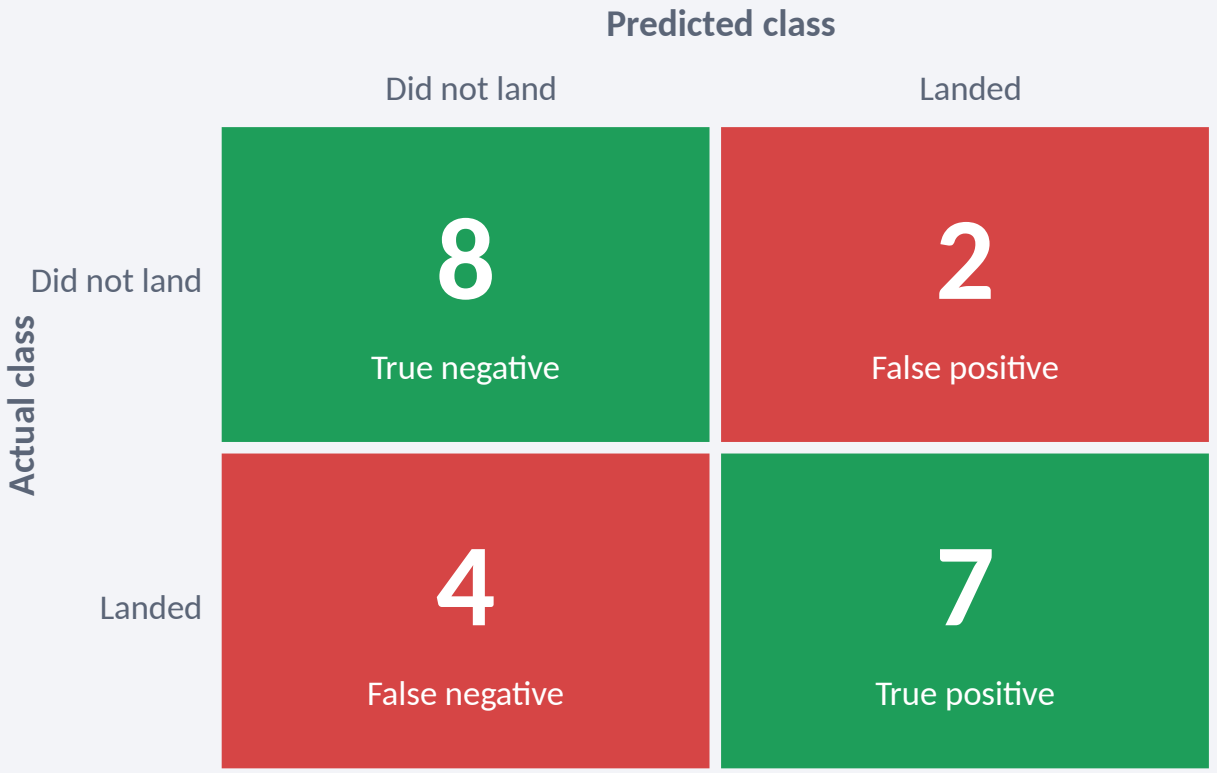


What it shows

- Decision Tree: best cross-validated accuracy (79%)
- Decision Tree and KNN tie on the test set (71%)
- Test set is only 21 launches, so one launch moves accuracy by about 5 points

80/20 split (random_state=2), 15 features, GridSearchCV with 10 folds

Confusion Matrix: Decision Tree

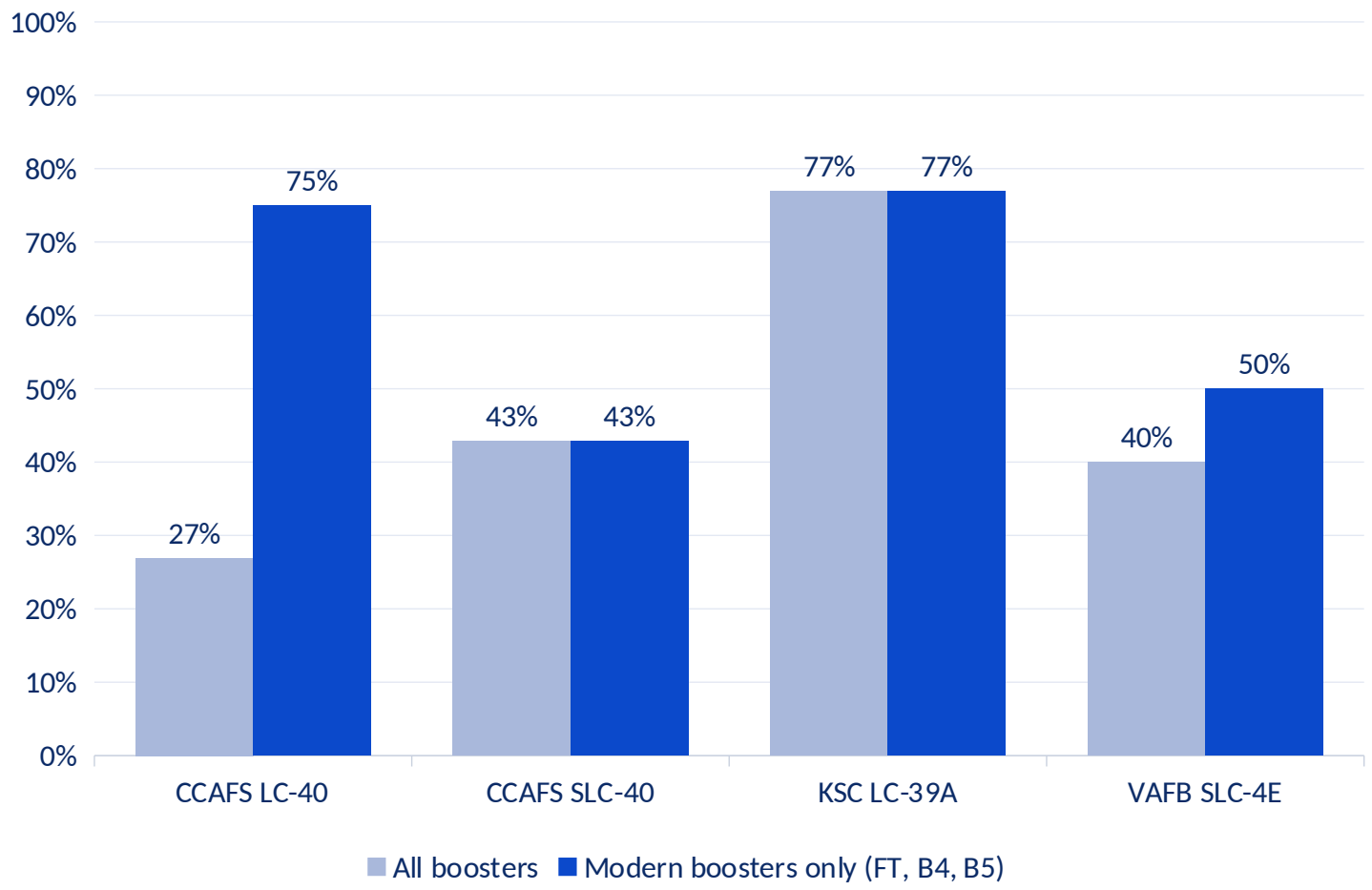


Reading the matrix

- Accuracy 71.4% (15 of 21 correct)
- Precision 78%: most predicted landings are real
- Recall 64%: it misses 4 real landings
- Errors lean cautious: more missed landings than false alarms

KNN produced the identical confusion matrix on this split

Insight 1: It Is the Booster, Not the Pad

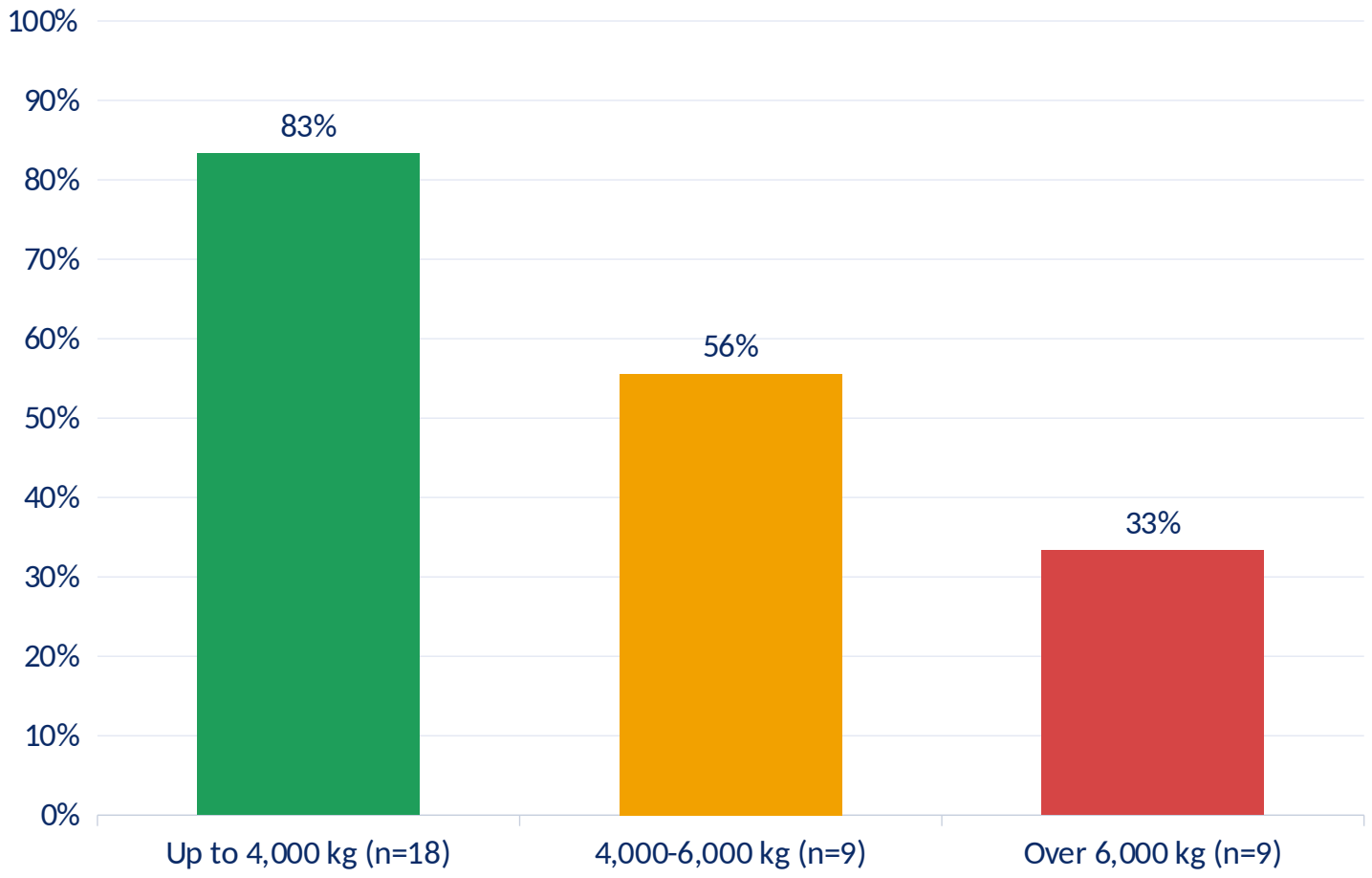


Why it matters

- CCAFS LC-40 looks weak (27%) because it flew the old boosters
- With modern boosters only it reaches 75%, level with KSC LC-39A (77%)
- Old boosters (v1.0, v1.1) landed 1 of 20; modern ones 23 of 36

Computed from spacex_launch_dash.csv; confounding by booster era

Insight 2: Payload Headroom Drives Landing Odds



Why it matters

- Modern boosters: 83% success up to 4,000 kg
- Falls to 33% above 6,000 kg
- Heavy missions leave less fuel for recovery, and some skip landing by design

Modern boosters only (FT, B4, B5), spacex_launch_dash.csv; class 0 includes no-attempt launches

Insight 3: Pricing Playbook for Space Y

Mission profile	Landing odds	Bid posture
Modern booster, up to 4,000 kg, LEO or SSO	High (about 80%+)	Price near reuse economics
Modern booster, 4,000-6,000 kg	Moderate (about 56%)	Add a recovery-risk margin
Heavy, over 6,000 kg, or GTO	Low (about 33-50%)	Assume expendable pricing

$103M

headroom per reused stage ($165M minus $62M)

62%

of a competitor price saved by reuse

Odds come from the landing data in this deck; treat the posture column as a starting point, not a quote.

Conclusions

1	**Learning curve**	Landing success rose from 0% (2010-14) to 88% (2020) as boosters matured.
2	**Booster generation**	FT, B4 and B5 boosters land 64% of the time versus 5% for v1.0/v1.1.
3	**Site and payload**	KSC LC-39A leads (77%). Payloads under 4,000 kg land far more often.
4	**Best model**	Decision Tree: CV 79%, test 71%. Promising, but the test set is tiny.
5	**Next steps**	Add landing-pad type, grid fins and reuse count; use time-based validation.

Appendix

SQL: ranking landing outcomes

```
SELECT Landing_Outcome, COUNT(*) AS n
FROM SPACEXTBL
WHERE Date BETWEEN "2010-06-04"
  AND "2017-03-20"
GROUP BY Landing_Outcome ORDER BY n DESC;
```

Python: model tuning

```
gs = GridSearchCV(model, params, cv=10)
gs.fit(X_train, y_train)
print(gs.best_params_, gs.best_score_)
gs.score(X_test, y_test)
```

Repository and tools

Repository: github.com/<your-username>/<your-repo>

- SpaceX API, requests, BeautifulSoup
- pandas, NumPy, SQLite
- Matplotlib, Seaborn, Folium, Plotly Dash
- scikit-learn (LR, SVM, Tree, KNN)
- Datasets: dataset_part_1/2/3.csv, spacex_launch_dash.csv

Thank you!

